	GLOBAL ECONOMICS **GRADE: 10TH** **LEARNING EVIDENCE 1** **TEACHER'S NAME: Nicolás López Cuéllar**	**3 TERM** **2025–2026**

Learning objective: Describes probabilities in continuous variables as a density function.	**Assessment criteria** C1. Describes probabilities in continuous variables as a density function. C2. Explains how a uniform random variable works. C3. Expresses how to work with distributions that have linear behaviour.

Criteria assessment

This activity assesses the three criteria of the learning evidence. Problems 1 to 8 focus mainly on criterion C1, problems 9 to 14 focus mainly on criterion C2, and problems 15 to 17 focus mainly on criterion C3. Criteria C1 and C2 each have a total of 33 possible points, while criterion C3 has a total of 35 possible points. A criterion is considered passed if the student earns at least 11 points in that criterion. The number of points available for each problem appears at the beginning of the task. Partial credit will be awarded when the procedure is correct but there are arithmetic errors. If the procedure is incorrect, no points will be awarded. Answers without procedure will not receive points.

1. First-Investment Product Resale Plans (C1)

An entrepreneur is comparing four product resale plans. The required initial investments are 10, 20, 30, and 40 million pesos. For each plan, the end-of-quarter value is modeled as uniformly distributed on a projected interval.

For the first plan, the final value is between 8 and 16 million pesos. For the second plan, it is between 16 and 28 million pesos. For the third plan, it is between 24 and 36 million pesos. For the fourth plan, it is between 30 and 45 million pesos.

A *loss* occurs when the final value is below the amount invested, and an *earning* occurs when the final value is above it.

Questions/Tasks.

a) **(2)** Write the density constant for each investment plan.

b) **(3)** Find the probability of loss and the probability of earning for each investment plan. **(0.5)** Then identify which investment has the highest probability of generating losses and which investment has the highest probability of generating earnings.

2. Online Coffee Sales Investment (C1)

A small entrepreneur runs a specialty coffee resale business. Let V be the end-of-quarter value, measured in million pesos. The final value cannot be below 30 and cannot exceed 60. Lower values are possible, but higher values become gradually more likely across this interval, so the final value is modeled with an increasing density on $[30, 60]$. The initial investment is 40 million pesos. The probability of the extreme value of 30 million pesos can be considered to be 0.

Questions/Tasks.

a) **(3)** Determine the density constant and write the pdf.

b) **(2)** Find the probability of loss and the probability of earning.

c) **(4)** Compute the probability that the money obtained falls between 40 and 50 million.

d) **(2)** Compute the probability that the money obtained is above 50 million.

3. Checking Three Candidate Density Functions I (C1)

A continuous random variable is modeled on the interval shown in each case.

Distribution A:	$f(x) = 0.20$	for $0 \le x \le 5$
Distribution B:	$f(x) = 0.30$	for $0 \le x \le 5$
Distribution C:	$f(x) = -0.20$	for $0 \le x \le 5$

Questions/Tasks.

a) (0.5) Decide which distribution is a valid PDF and which ones are not. Justify your answer.

b) (0.5) For the valid distribution, find $P(1 \le X \le 4)$.

4. Checking Two Candidate Density Functions II (C1)

A continuous random variable is modeled on the interval shown in each case.

Distribution A	$f(x) = \dfrac{x}{18}$	for $0 \le x \le 6$
Distribution A	$f(x) = \dfrac{x}{12}$	for $0 \le x \le 6$

Questions/Tasks.

a) (1) Decide which distribution is a valid PDF and which one is not. Justify your answer.

b) (1) For the valid distribution, find $P(3 \le X \le 6)$.

5. Checking Two Candidate Density Functions III (C1)

A continuous random variable is modeled on the interval shown in each case.

Distribution A	$f(x) = \dfrac{6-x}{12}$	for $1 \le x \le 5$
Distribution B	$f(x) = \dfrac{6-x}{10}$	for $1 \le x \le 5$

Questions/Tasks.

a) (2) Decide which distribution is a valid PDF and which one is not.

b) (2) For the valid distribution, find $P(1 \le X \le 3)$.

6. Battery Charging Time (C1)

Let X be the battery charging time (in h) with support $[1, 5]$ and density

$$f(x) = k \quad \text{for } 1 \le x \le 5.$$

Questions/Tasks.

a) (1) Find k so $f(x)$ is a valid PDF.

b) (1) Find $P(2 \le X \le 4)$.

7. Startup Delay (C1)

Let T be a startup delay (in s) with support $[0, 8]$ and density

$$f(t) = kt \quad \text{for } 0 \le t \le 8.$$

Questions/Tasks.

a) (1) Find k so $f(t)$ is a valid PDF.

b) (2) Find $P(T \ge 6)$.

8. Packing Score (C1)

Let Q be a packing score with support $[2, 6]$ and density

$$f(q) = k(q-1) \quad \text{for } 2 \le q \le 6.$$

Questions/Tasks.

a) (3) Find k so $f(q)$ is a valid PDF.

b) (2) Find $P(3 \le Q \le 5)$.

9. Two-Scenario Inventory Investment (C2)

A seller invests in a short-term inventory deal to resell packaged foods in nearby cities. For the quarter, they consider two realistic market scenarios. In Scenario 1, transport bottlenecks continue and restocking stays slow; this has probability 0.4. In Scenario 2, transport returns to normal and inventory rotates faster; this has probability 0.6. If Scenario 1 occurs, the final value of the investment is expected to end between fifty and seventy million pesos, with all values in that interval treated as equally plausible. If Scenario 2 occurs, the final value is expected to end between sixty and ninety million pesos, again with all values in that interval treated as equally plausible. The initial investment is sixty-five million pesos. A loss occurs if the final value is below that amount, and an earning occurs if the final value is above it.

Questions/Tasks.

- **a) (2)** Write the conditional density for each scenario.
- **b) (3)** Build the unconditional pdf of the final value as a piecewise function.
- **c) (2)** Verify that the unconditional pdf integrates to one.
- **d) (3)** Find the probability of loss and the probability of earning.
- **e) (2)** Find the probability that the final value falls inside the overlap interval.

10. Uniform Distribution I (C2)

Let $X \sim \text{Uniform}(1, b)$, where $b > 8$.

Questions/Tasks.

- **a) (1)** Find the density value $f(x)$ on the support.
- **b) (1)** Find $P(2 \le X \le 7)$.
- **c) (1)** Find $P(2 \le X \le 7)$ when $b = 9$ and $b = 16$.

11. Uniform Distribution II (C2)

Let X be the delivery time (in days) for a certain shipping route. Assume

$$X \sim \text{Uniform}(a, b),$$

where $a < 3 < 6 < b$. Here, a is the earliest possible delivery time and b is the latest possible delivery time.

Questions/Tasks.

- **a) (2)** Find the density value $f(x)$ on the support.
- **b) (2)** Find $P(3 \le X \le 6)$.
- **c) (1)** Find $P(3 \le X \le 6)$ when $a = 1, b = 9$, $a = 1, b = 16$, $a = 1, b = 10$, and $a = -6, b = 9$.

12. Two-Interval Piecewise Uniform (C2)

Let X be the final value of a small short-term investment, measured in million pesos. The investor estimates that lower outcomes between 0 and 4 are more likely, while higher outcomes between 4 and 10 are less likely. The final value is modeled by the density

$$f(x) = \begin{cases} \frac{1}{5}, & 0 \le x < 4, \\ \frac{1}{30}, & 4 \le x \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- **a) (1)** Find $P(X < 3)$.
- **b) (1)** Find $P(X \ge 3)$.
- **c) (2)** Find $P(2 \le X \le 8)$.

13. Two-Interval Piecewise Uniform with Unique Variable Endpoint (C2)

Let X have density

$$f(x) = \begin{cases} \frac{1}{12}, & -2 \le x < 1, \\ \frac{1}{8}, & 1 \le x \le b, \\ 0, & \text{otherwise,} \end{cases}$$

where $b > 1$.

Questions/Tasks.

- **a) (2)** Find the value of b so this is a valid probability distribution function.
- **b) (1)** Find $P(X < -1)$.
- **c) (1)** Find $P(X > 3)$.
- **d) (1)** Find $P(-1 \le X \le 3)$.

14. Piecewise Uniform with Unknown k (C2)

Let X have density

$$f(x) = \begin{cases} 2k, & 0 \le x < 2, \\ k, & 2 \le x \le 6, \\ 0, & \text{otherwise}, \end{cases}$$

where $k > 0$.

Questions/Tasks.

- **a) (2)** Find the value of k so that f is a valid PDF.
- **b) (1)** Find $P(X < 4)$.
- **c) (1)** Find $P(4 \le X \le 6)$.

15. Small-Business Investment (C3)

An investor invests in a short-term small-business opportunity: buying low-cost clothing bundles to resell online and at weekend markets. They estimate the final value cannot be below zero million pesos because, in this simplified model, the end-of-period value cannot be negative. They also estimate it cannot exceed thirty million pesos because their first-round sales capacity is limited. Their most likely final value is twelve million pesos, based on expected demand from regular customers. Values below twelve million pesos are possible, but become less likely as results move farther down from that level. Values above twelve million pesos are also possible, but after that point, very high outcomes become gradually less likely. Since the model is continuous, the probability of obtaining any exact extreme value can be considered 0, so the probabilities of ending at exactly zero million pesos or exactly thirty million pesos are treated as 0. The initial investment is six million pesos. A loss occurs if the final value is below that amount, and an earning occurs if the final value is above it.

Questions/Tasks.

- **a) (4)** Write the pdf as a piecewise function.
- **b) (1)** Verify that the total area under the density is one.
- **c) (3)** Find the probability of loss and the probability of earning.
- **d) (4)** Find the specific probability of earning more than 50% of the investment.

16. Basic Symmetric Triangular Density (C3)

A random variable Y is modeled on the support $[0, 8]$ by one of the following two functions:

$$f(y) = \begin{cases} \dfrac{y}{16}, & 0 \le y \le 4, \\ \dfrac{8-y}{16}, & 4 \le y \le 8, \\ 0, & \text{otherwise}, \end{cases}$$

$$g(y) = \begin{cases} \dfrac{y}{12}, & 0 \le y \le 4, \\ \dfrac{8-y}{12}, & 4 \le y \le 8, \\ 0, & \text{otherwise}. \end{cases}$$

Questions/Tasks.

- (a) **(3)** Decide which of the two functions is a valid probability density function on the interval $[0, 8]$. Justify your answer.
- (b) **(4)** For the valid density, compute $P(Y > 2)$.
- (c) **(4)** For the valid density, find the most probable interval of length 2, compute its probability, and draw the corresponding graph.

17. Linear Density with variable endpoint (C3)

Let X be a continuous random variable with density

$$f(x) = kx, \quad 0 \le x \le b,$$

and $f(x) = 0$ otherwise, where $b > 0$ and $k > 0$.

Questions/Tasks.

- (a) **(3)** Find k geometrically so that f is a valid PDF, and draw the corresponding graph.
- (b) **(3)** Compute $P(0 \le X \le a)$, where $0 \le a \le b$, and draw the corresponding graph.
- (c) **(3)** Compute $P(a \le X \le b)$, where $0 \le a \le b$, and draw the corresponding graph.
- (d) **(3)** Compute $P(a \le X \le c)$, where $0 \le a \le c \le b$, and draw the corresponding graph.